

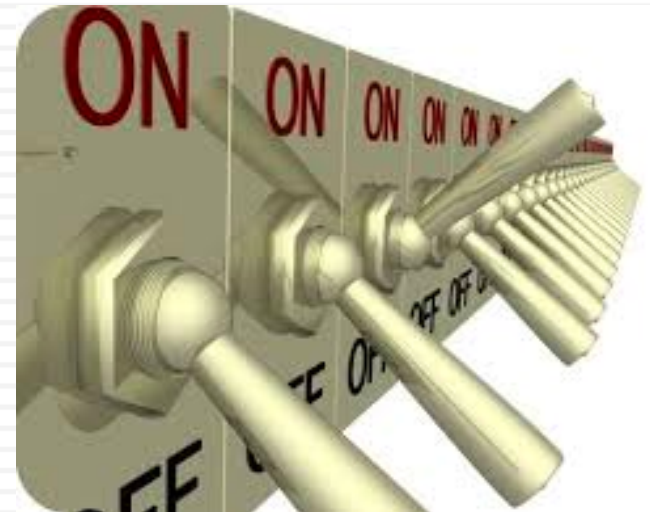
UNIT 4: MOLECULAR GENETICS

4.4 – GENE EXPRESSION

Biology 9 Mr. Queenan
Ch. 14.3 (p. 451-456)

Gene Expression/Regulation Outline

1. Prokaryotic Regulation
2. Eukaryotic Regulation
 1. Chromatin
 2. Transcriptional Control
 3. Post-Transcriptional Control
 4. Translational Control
 5. Post-Translational Control
3. Mutations



[Amoeba Sisters - Gene Regulation](#)

Regulation

- What is meant by gene regulation?
 - ▣ The process of turning genes on and off
- Why is it necessary for genes to be regulated?
 - ▣ All genes do not need to be transcribed and translated at all times. By regulating genes, gene products can be produced only when and if needed by a cell or organism.
- Regulation differs in prokaryotes and eukaryotes



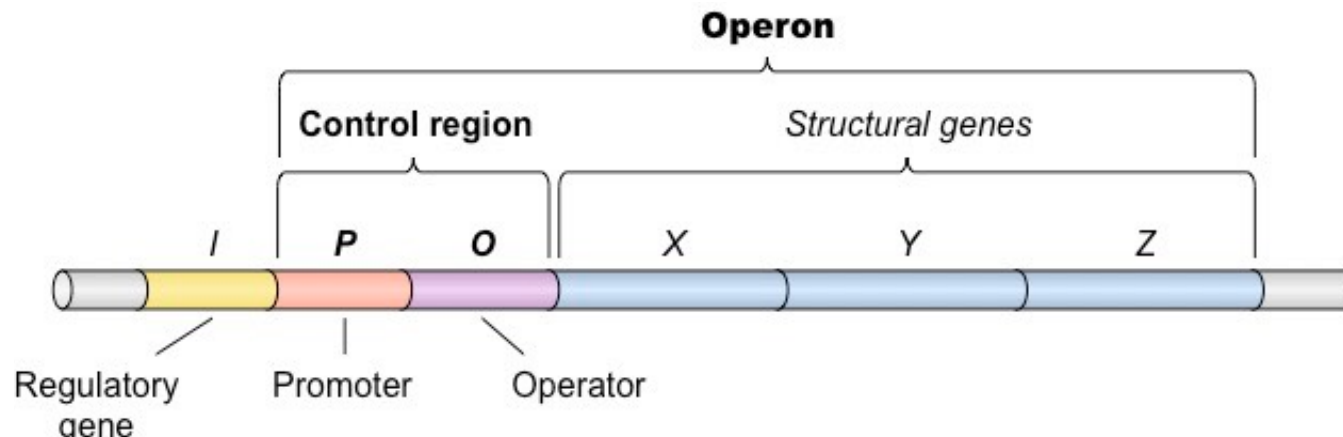
Prokaryotic Regulation

Prokaryotic Regulation

- Prokaryotic DNA is normally less complex than eukaryotic
 - ▣ Genes for a certain pathway are all grouped together so that they are transcribed at the same time
 - ▣ These are termed **Structural Genes**
- Gene regulation in prokaryotes controlled by repressors, and the entire gene encoding region known as an **Operon**
- Some operons in always on state unless regulated, and others in always off unless regulated

Operons

- The operon contains the following:
 - ▣ **Regulator gene** - Located outside the operon. Codes for DNA binding proteins that act as repressors.
 - Controls if the operon is active or not
 - ▣ **Promoter** – Short sequence of DNA found before the structural genes. RNA polymerase binds to the promoter. Binding signals start of transcription
 - ▣ **Operator** – Portion of DNA following promoter where a repressor will bind. When bound, polymerase cannot bind to promoter
 - ▣ **Structural genes**– Genes that code for enzymes and proteins

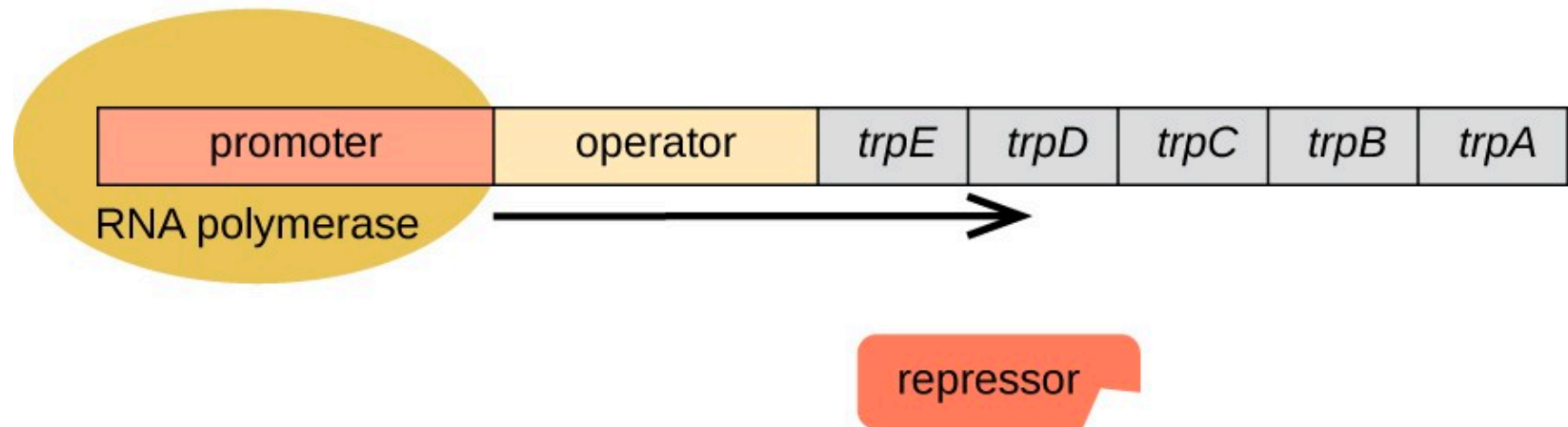


trp Operon

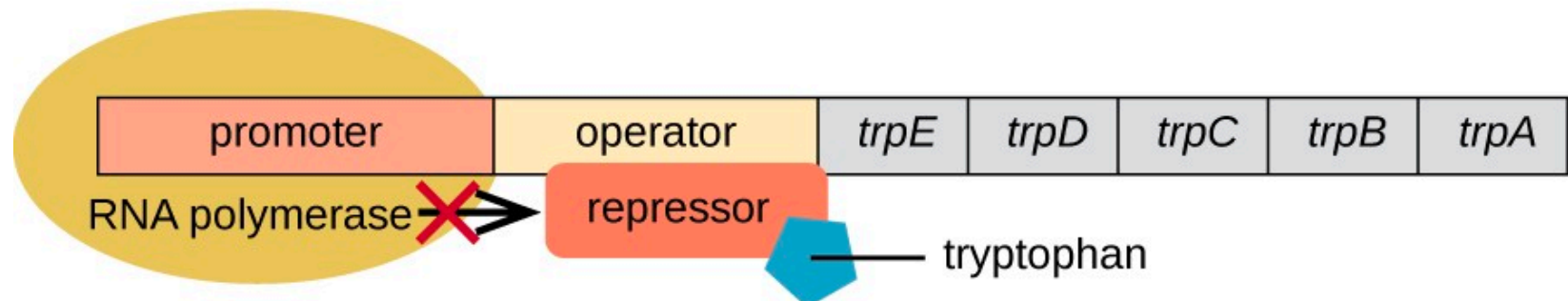
- Always "on" operon unless regulated
 - ▣ Known as a **repressible operon**
 - ▣ Repressible operons usually associated with *anabolic processes*
- Structural genes in this operon code for 5 enzymes used in synthesis of tryptophan for use by the cell
- When no tryptophan present, genes are transcribed and enzymes to make tryptophan created.
- If tryptophan present, enzymes are not needed
 - ▣ Tryptophan binds to the repressor protein
 - ▣ Repressor changes shape due to binding, and can now bind to the operator, blocking the promoter
 - ▣ Structural genes not expressed
 - Tryptophan is a **co-repressor**

trp Operon

In the absence of tryptophan, the *trp* repressor dissociates from the operator, and RNA synthesis proceeds.



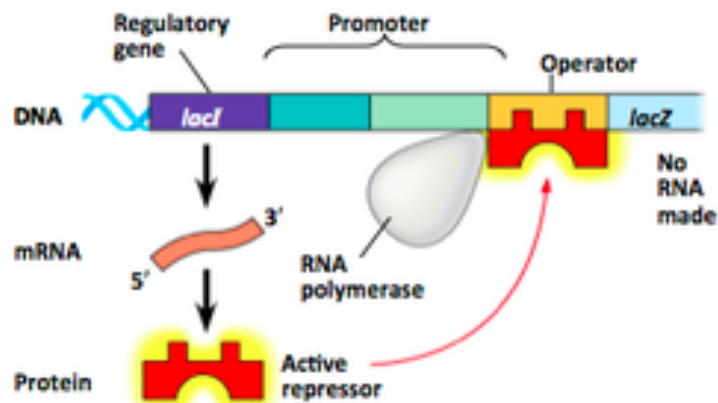
When tryptophan is present, the *trp* repressor binds the operator, and RNA synthesis is blocked.



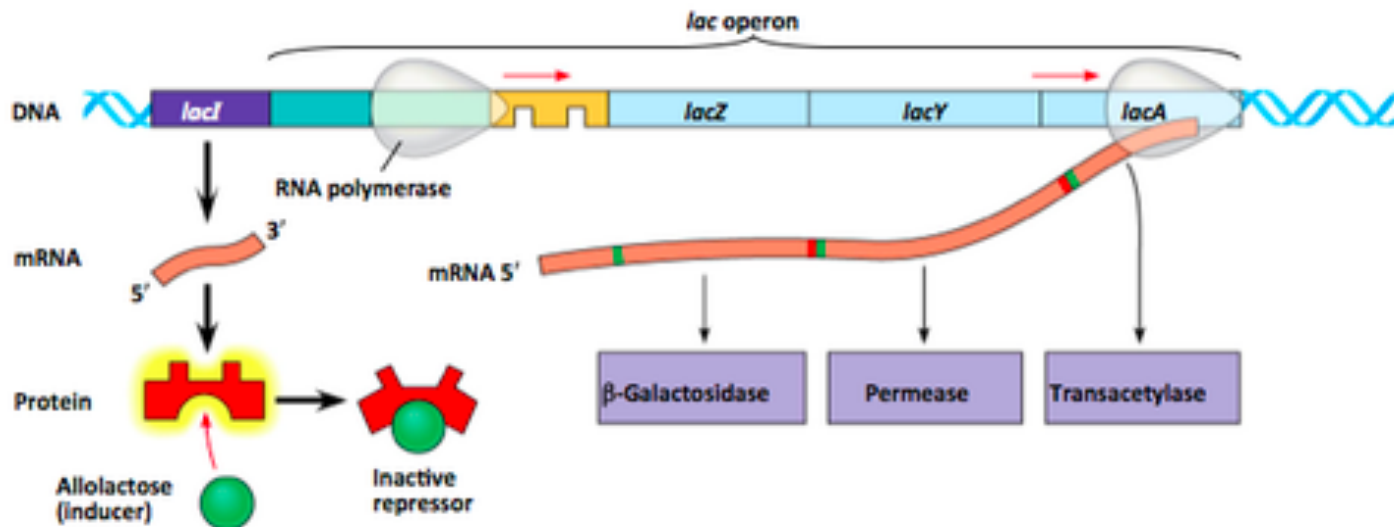
lac Operon

- *lac* Operon normally off, unless regulated
 - ▣ Known as an **inducible operon**
 - ▣ Lactose is an **inducer** of expression of the operon
 - ▣ Inducible operons usually associated with *catabolic operon*
- No need to produce the enzymes to break down lactose into glucose when lactose not present in the environment
- When no lactose present, repressor binds to the promoter to stop expression of *lac* operon
- When lactose present
 - ▣ Lactose binds to repressor, causing it to change shape and fall off
 - ▣ Leads to promoter being bound by RNA Polymerase
 - ▣ Production of Beta-galactosidase, a permease, and transacetylase, all involved in lactose metabolism

lac Operon



(a) Lactose absent, repressor active, operon off



(b) Lactose present, repressor inactive, operon on

Expression of the *lac* operon requires induction



Eukaryotic Regulation

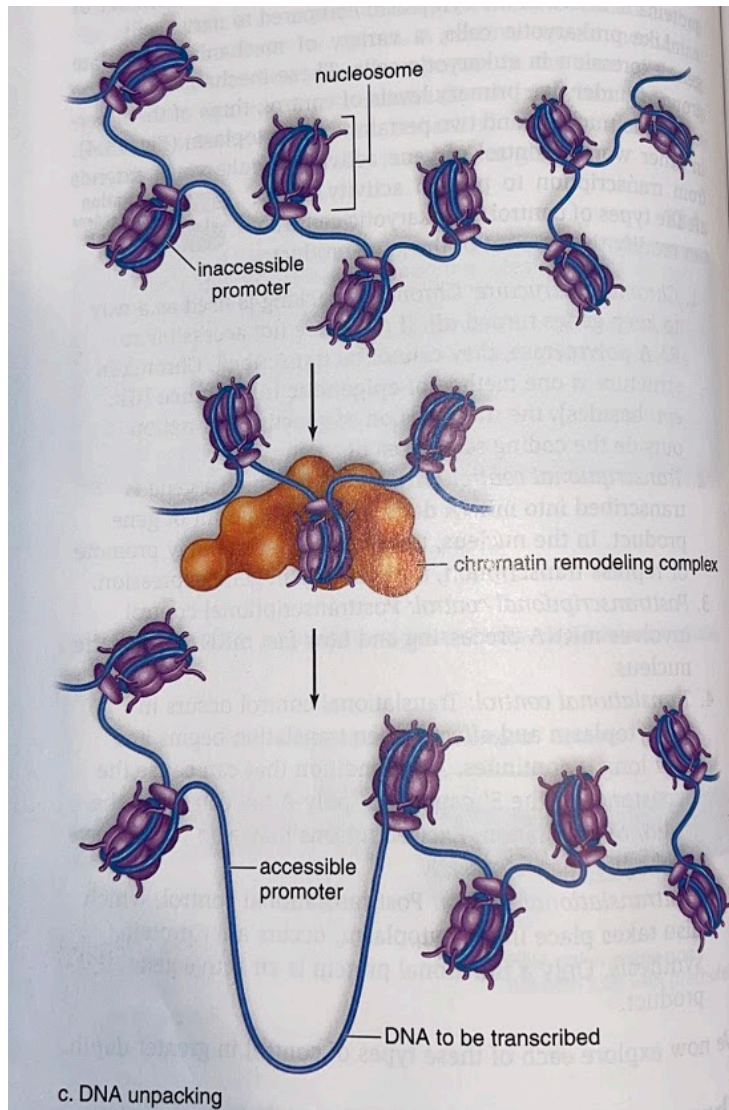
Eukaryotic Regulation

- With few exceptions, every cell of a eukaryote has a full complement of chromosomes and DNA
 - Difference in cell type determined by the different genes that are active and expressed in a cell
 - 5 levels of control in eukaryotic cells
 - 1. Chromatin
 - 2. Transcriptional Control
 - 3. Post-Transcriptional Control
 - 4. Translational Control
 - 5. Post-Translational Control
- } Regulation in Nucleus
- } Regulation in Cytoplasm

1) Chromatin Structure Regulation

- Euchromatin is DNA less compacted and available for transcription. Heterochromatin more compacted and not going through transcription.
- Histone proteins that are involved in compaction of DNA are also involved in regulation.
 - Histones have tails
 - In heterochromatin, tails have methyl group added, or are **methylated** ($-\text{CH}_3$)
 - In euchromatin, tails have acetyl groups added, or are **acetylated** ($-\text{COCH}_3$)
 - **Chromatin Remodeling Complex** can push aside acetylated histones to allow transcription to occur

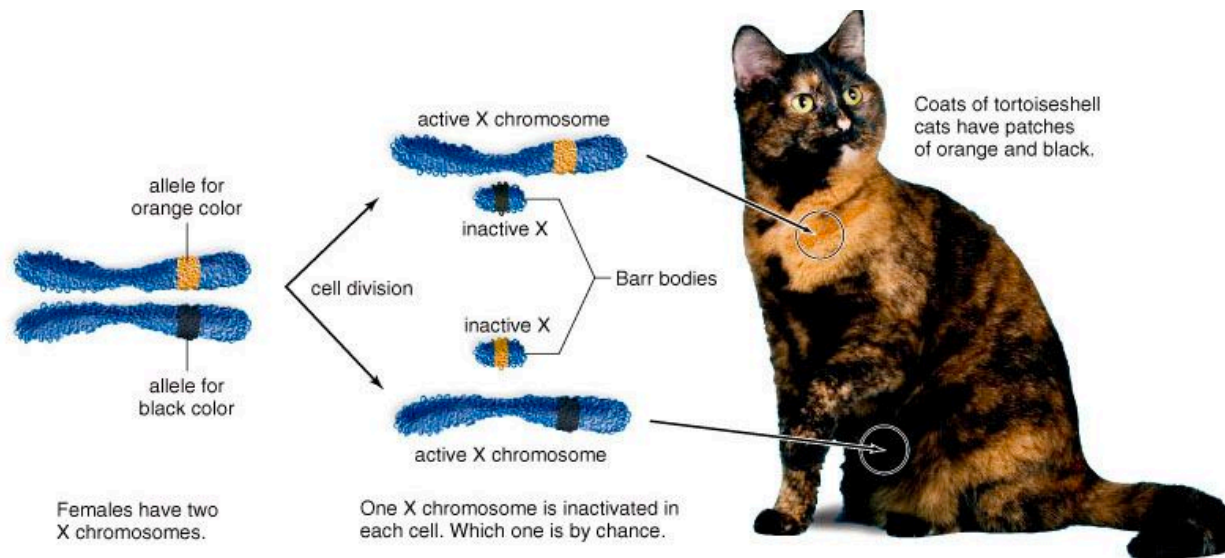
1) Chromatin Structure Regulation



Chromatin Remodeling Complexes can also add methyl or acetyl groups onto histones for further regulation of DNA regions

1) Chromatin Structure Regulation

- Heterochromatin is not transcribed
 - In genetic females, one X chromosome is inactivated in a process called X-inactivation – creates a **Barr Body**
 - One gets highly condensed, and unavailable for transcription
 - Means genetic males and females will produce same amount of gene products from X chromosome
 - X chromosome inactivated differs from cell to cell
 - Can result in mosaic expression with some heterozygous genes



1) Chromatin Structure Regulation

□ **Epigenetic Inheritance**

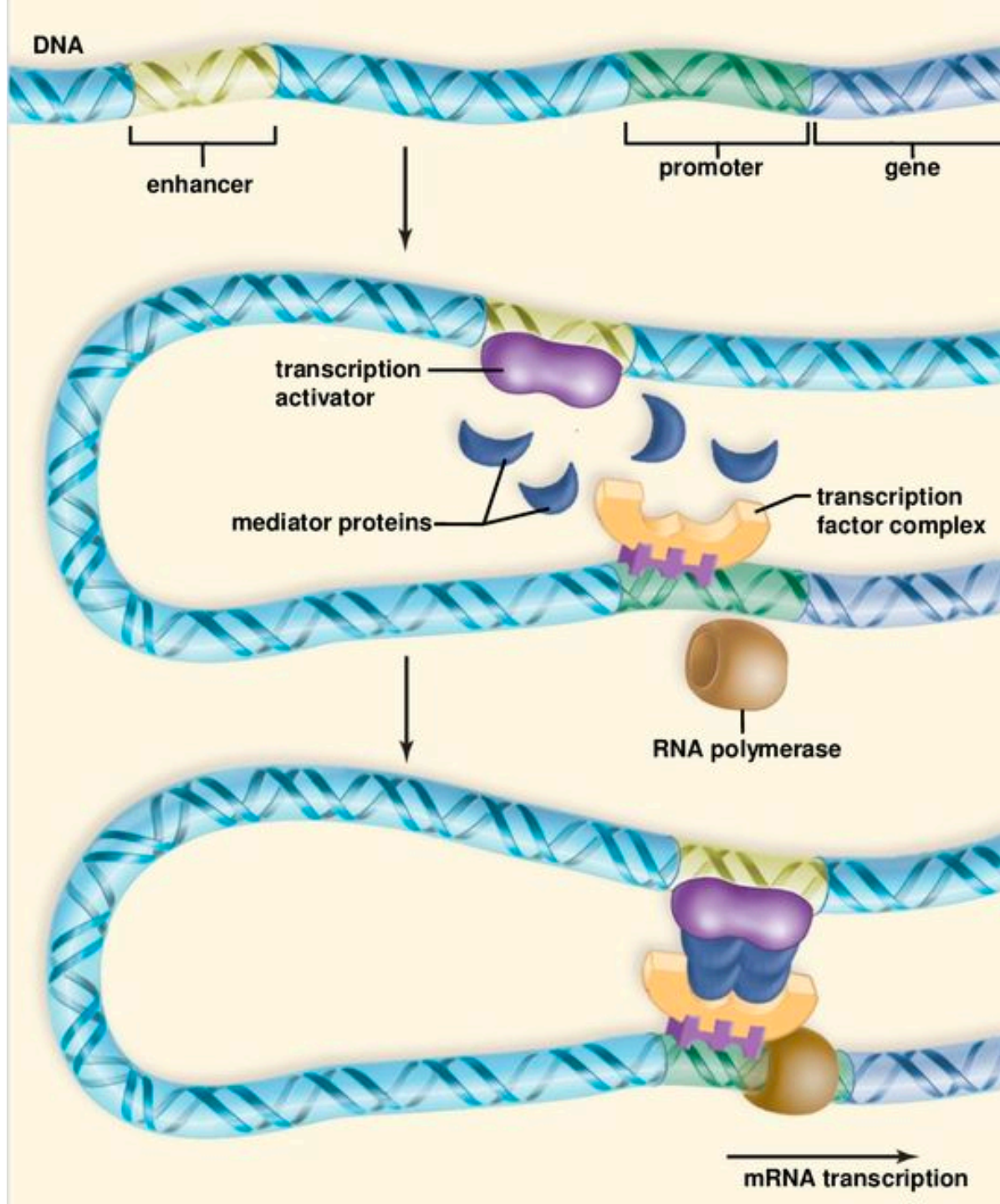
- Variations in a pattern of inheritance not due to the DNA but instead, due to methylation
 - Histones for a gene get methylated, and as a result, DNA itself becomes methylated as well
 - Stops expression of that DNA

□ **Genomic Imprinting**

- Process of mother or father's genes (but not both) being methylated during gamete formation.
- If inherited allele for a gene is methylated, it is not expressed unless an unmethylated version is inherited from the second parent
- Genes are normal in all regards except for methylation

2) Transcriptional Control

- Most critical level of eukaryotic regulation
- Transcriptional control much more complicated in eukaryotes compared to prokaryotes
- Eukaryotic cells do not contain operons
 - Instead every cell has different types of **transcription factors** that help regulate transcription.
 - Assist in binding of RNA polymerase to a promoter, and some promoters require several T-factors for transcription to occur
 - Some promoters will also require a **transcription activator** to assist prior to transcription
 - **Enhancers** located elsewhere on the chromosome may need to be bound by factors in order for a structural change to occur and begin transcription
 - **Repressors** may also bind to **silencer** regions of the promoter of a gene and stop transcription

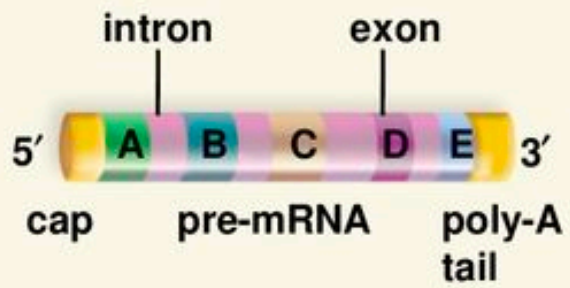


2) Transcriptional Control

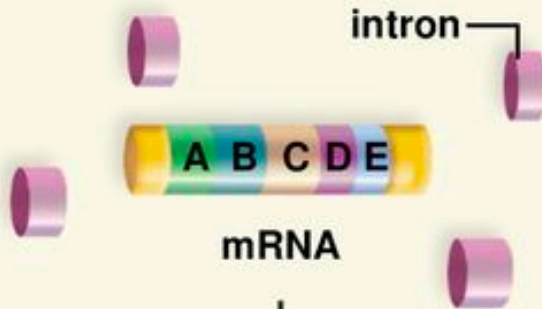
- Transcription factors, activators, enhancers and repressors all always found within the nucleus
- Must be activated to do their job
 - ▣ **Kinases** – Phosphorylate these molecules to activate their activity
 - ▣ **Phosphatases** – Dephosphorylate the molecules
 - Together these play a role in cell signaling network for cell growth

3) Post-Transcriptional Control

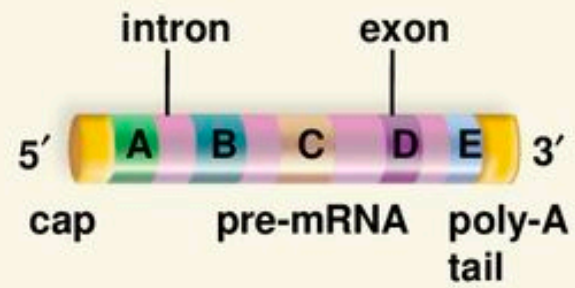
- Relates to alternative splicing of mRNA, and speed of movement out of the nucleus
- **Alternative Splicing**
 - After transcription, intron regions are removed and exons are combined to form the template for the protein to be created
 - Differential splicing of the introns can occur, resulting in alternative forms of the final mRNA product.
 - Allows DNA to code for many more protein products than a strict reading of the DNA would suggest
 - Sometimes, an intron may be left in the final mRNA product (known as small nuclear, or snRNA). Results in destruction of mRNA before leaving nucleus



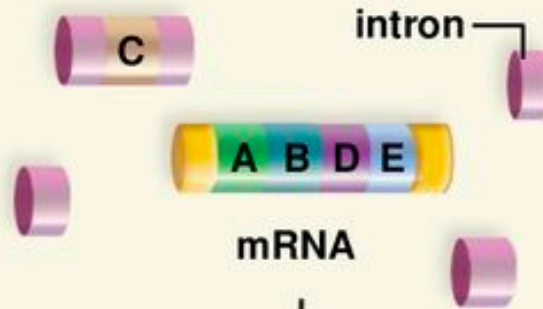
RNA splicing



protein product 1



RNA splicing



protein product 2

3) Post-Transcriptional Control

□ **Small RNA (sRNA)**

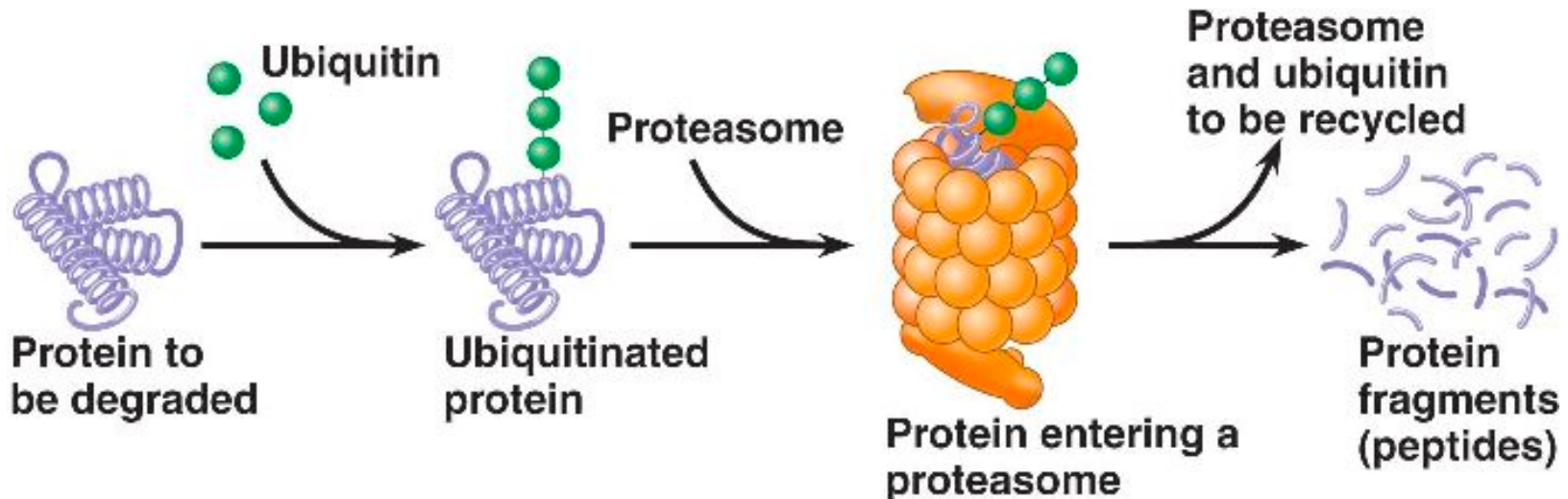
- Only ~ 1.5% of all DNA in a cell encodes for genes
- For longest time, remaining 98.5% was considered “junk”
- Now know that this region contains sRNA, involved with regulation
 - Can alter the compaction of DNA so genes are not accessible
 - Source of **microRNAs (miRNAs)**.
 - Small pieces of DNA that bind to and dampen translation in the cytoplasm
 - Source of **small-interfering RNA (siRNAs)**
 - Join with an enzyme to silence mRNA products from being translated
 - miRNA & siRNA now used in gene therapy. Won Nobel Prize 2006

4) Translational Control

- Occurs in the cytoplasm after mRNA leaves nucleus
- Determines degree to which mRNA is translated into a protein product
 - Features of the mRNA affect whether translation occurs and how long the mRNA remain active
 - Presence of 5' cap
 - Length of poly-A tail on 3' end
 - miRNA may regulate these products by causing their destruction before translation

5) Post-translational Control

- Affects the activity of the protein product
 - ▣ Accomplished by regulating activation and degradation rate
 - **Activation** – Some proteins are not immediately active after synthesis
 - A protein may fold into 3-D conformation and then need to be cleaved into parts before becoming its active form
 - **Degradation – Proteases** will degrade proteins. Proteases are confined to areas known as proteasomes, a special type of lysosome.
 - A protein gets tagged by signaling protein that signifies it should be moved to the proteasome for degradation





Mutations

Gene Mutations

- A gene mutation is a permanent change in the sequence of bases in DNA
- The effects of a gene mutation can range from
 - ▣ No effect on protein activity to
 - ▣ Complete inactivation of the protein
- Germ-line mutations occur in sex cells
- Somatic mutations occur in body cells

Gene Mutations

□ Spontaneous Mutations

- ▣ Chemical changes in DNA that lead to mispairing during replication
- ▣ Movement of transposons from one chromosomal location to another
- ▣ Replication Errors
 - DNA polymerase
 - Proofreads new strands
 - Generally corrects errors
 - Overall mutation rate is 1 in 1,000,000,000 nucleotide pairs replicated

□ Induced mutations

- ▣ Caused by **mutagens** such as radiation and organic chemicals
- ▣ Many mutagens are also carcinogens (cancer-causing)
- ▣ Environmental Mutagens
 - Ultraviolet Radiation
 - Tobacco Smoke

Gene Mutations

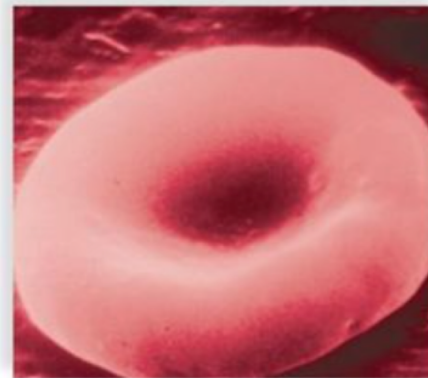
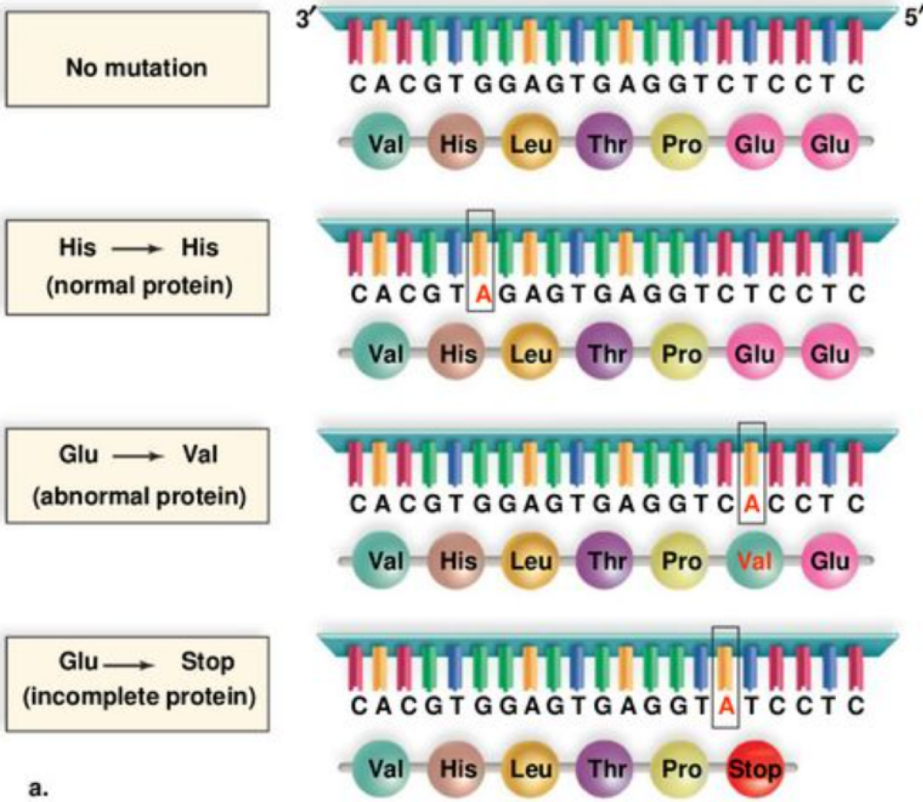
□ Point Mutations

- ▣ Involve a change in a single DNA nucleotide
- ▣ Change one codon to a different codon
- ▣ Effects on the protein vary:
 - Nonfunctional
 - Reduced functionality
 - Unaffected

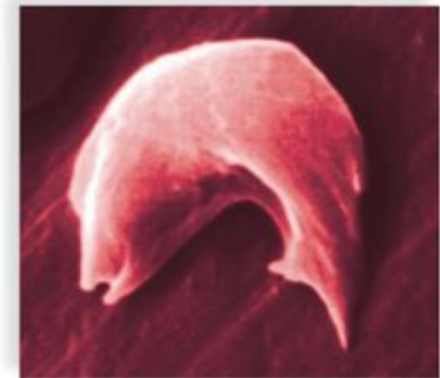
□ Frameshift Mutations

- ▣ One or two nucleotides are either inserted or deleted from DNA
- ▣ The protein is always rendered nonfunctional
 - Normal: THE CAT ATE THE RAT
 - After deletion: THE ATA TET HER AT
 - After insertion: THE CCA TAT ETH ERA T

Point Mutations in Hemoglobin



b. Normal red blood cell

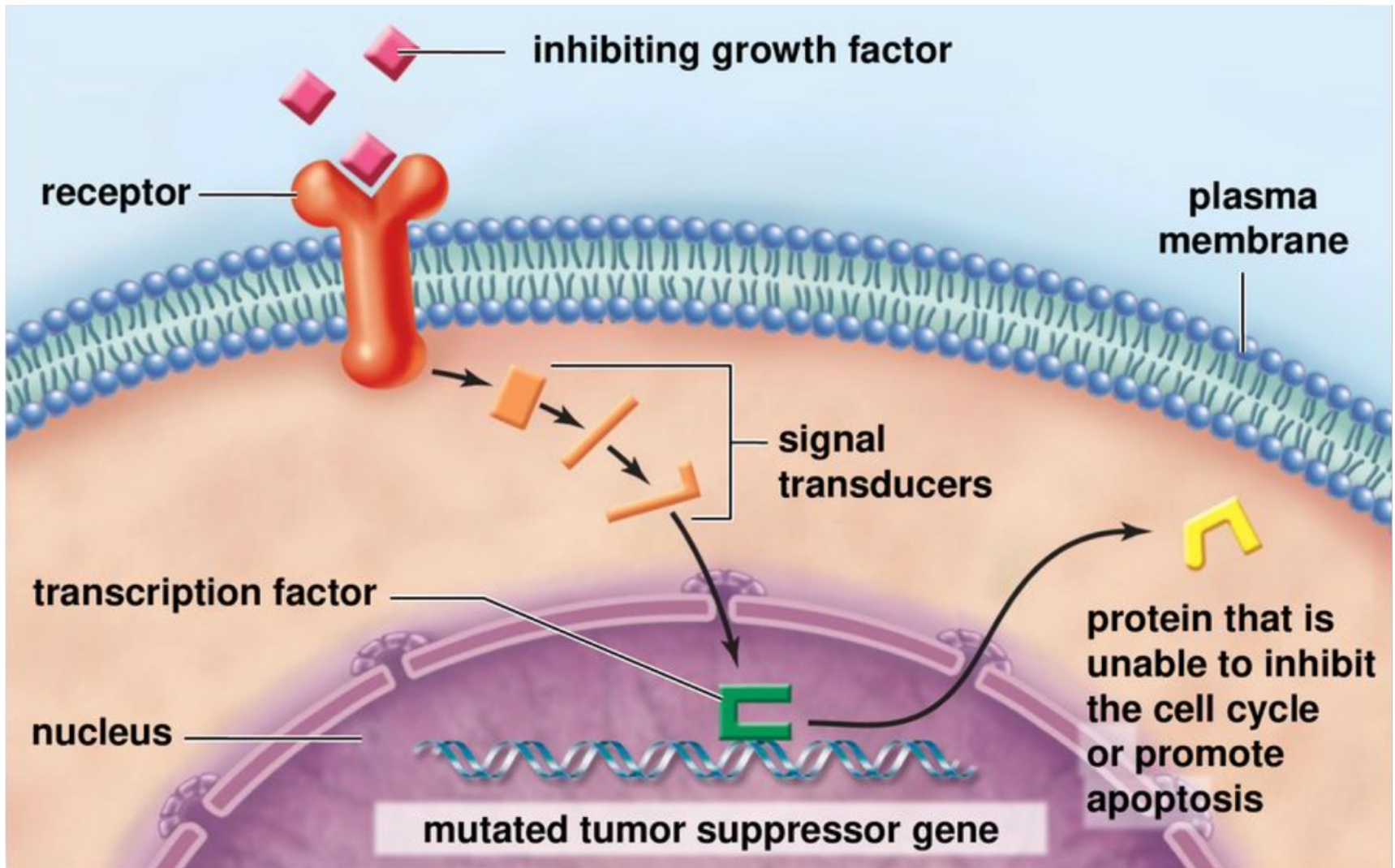


c. Sickled red blood cell

Gene Mutations

- Development of cancer involves a series of accumulating mutations
- Proto-oncogenes – Stimulate cell division
 - Mutated proto-oncogenes become oncogenes that are always active
- Tumor suppressor genes – inhibit cell division
- Mutations in oncogene and tumor suppressor genes:
 - Stimulate the cell cycle uncontrollably
 - Lead to tumor formation

Cell Signaling Pathway That Stimulates a Mutated Tumor Suppressor Gene



Cell Signaling Pathway That Stimulates an Oncogene

